

Subsystem Specification Recovery & signalling

Make the boat easy to see, easy to find and easy to retrieve with no power, no software and no radio.

Document	BOATY-SSS-REC
Issue	Issue B (baselined, sdr-baseline-1)
Date	30 September 2026
Status	Baselined at the System Design Review (tag sdr-baseline-1); changes only by change request
Parents	SRS Issue H, ADD Issue H, ICD Issue H, FMEA Issue G, KCL Issue D
Content	6 allocated SRS requirements → 8 subsystem requirements (6 M, 1 S, 1 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	30 September 2026	SDR WP5 (RID-08): issue, parents and applicable documents taken from the baseline register; issued as the SDR baseline candidate.	Claude

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Make the boat easy to see, easy to find and easy to retrieve with no power, no software and no radio.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Hi-vis finish • Flag • Recovery hoop • LED beacon ring (hardware) • 'If found' labels • Optional tracker pocket • Bank recovery kit	• Beacon drive and patterns (HLM) • Mast (HUL)

1.3 Breakdown

Item	Description
REC-1 Visibility	Fluorescent parts/tape, flag
REC-2 Retrieval	Hoop, recovery kit
REC-3 Identification	Labels, tracker pocket
REC-4 Beacon	WS2812 ring at the mast head

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue H	Parent requirements
BOATY-ADD-001 Issue H	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue H	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue G	Failure modes and actions carried into this issue
BOATY-KCL-001 Issue D	Component data, parameters and cost

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-09 Beacon drive	Helm	Conforms	§6

4. Parent requirements

Primary (6): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
REC-002	M	At least 50% of the visible top surface shall be fluorescent orange or yellow.	REC-D01
REC-003	M	A flag at least 100 × 60 mm shall fly with its top at least 300 mm above the waterline.	REC-D02
REC-004	S	An LED beacon shall be visible from 100 m in overcast daylight, with flash patterns showing armed, failsafe and...	REC-D03
REC-005	M	A recovery hoop of at least 60 mm internal diameter shall be fitted to the mast. It shall withstand a 50 N pull, and...	REC-D06
REC-006	M	The hull shall carry an 'If found' label with a contact phone number.	REC-D04
REC-007	C	A Bluetooth item tracker may be carried in the hull.	REC-D05

5. Subsystem requirements

Requirements are numbered REC-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Visibility and identification

ID	Requirement	Pri	Ver	Stage	Trace
REC-D01	At least 50% of the visible top area shall be fluorescent orange or yellow, measured on a top-down photo.	M	T	BENCH	REC-002
REC-D02	A fluorescent ripstop flag $\geq 120 \times 80$ mm shall fly with its top ≥ 320 mm above the waterline, below the GNSS.	M	I	BENCH	REC-003, IF-19
REC-D03	The beacon ring shall be visible at 100 m in overcast daylight and show the helm's state patterns (IF-09).	S	T	LAKE	REC-004, IF-09
REC-D04	Both hulls shall carry a waterproof 'IF FOUND' label with the owner's phone number.	M	I	BENCH	REC-006
REC-D05	An optional sealed pocket in a mid segment shall take a ≤ 35 mm tracker tag.	C	I	BENCH	REC-007

5.2 Retrieval

ID	Requirement	Pri	Ver	Stage	Trace
REC-D06	A printed hoop (≥ 60 mm internal) integrated with the mast collar shall take a 50 N pull. The boat tows by it without capsizing.	M	T	POOL	REC-005
REC-D07	The bank kit shall contain a ≥ 3 m telescopic pole with net and a casting rod with a weighted float line, packed with the boat.	M	I	LAKE	OPS-008
REC-D08	The flag staff shall be capped, and hoop and flag edges rounded (≥ 0.6 mm).	M	I	BENCH	CHD-002

6. Design constraints and budgets

- Passive measures must not depend on power (SRS 5.14 intent)
- Child-safe finish (CHD-002)

Budget	Allocation
Mass allocation	≤ 45 g (estimate 40 g)
Cost allocation	£3 (LED ring, tape); kit from owned fishing gear
Power	Beacon ≤ 0.3 W average

7. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	1	-	4	-	-	5
POOL	1	-	-	-	-	1
LAKE	1	-	1	-	-	2

Ref	Open item
TBC-08	Beacon data level (IF-09)
REC-OI-1	Owner to supply the contact number for the labels